Practical 5A: Create an MVC Application to Perform Basic Arithmetic Operations

Steps

1. Open Visual Studio 2022/2026.

2. Create a new project.

- Click “Create a new project”.
- Search for MVC.
- Select “ASP.NET Core Web App (Model-View-Controller)”.
- Click Next.

2.2 Enter the project name as arithmeticmvc, click Next, and select the latest available framework.

3. Create Model File.

- In Solution Explorer (if it is not visible, open View → Solution Explorer).
- Right-click Models → Add → Class.
- Enter the file name Calculator.cs and click Add.

Calculator.cs — Model Code

```
namespace arithmeticmvc.Models
{
    public class Calculator
    {
        public double Number1 { get; set; }
        public double Number2 { get; set; }
        public required string operation { get; set; }
        public double Result { get; set; }
    }
}
```

Press Ctrl + S to save the file.

4. Open and Modify the Controller

In Solution Explorer, open Controllers → HomeController.cs. Replace the existing controller code with the following:

HomeController.cs

```
using arithmeticmvc.Models;
using Microsoft.AspNetCore.Mvc;
```

```
using System.Diagnostics;

namespace arithmeticmvc.Controllers
{
    public class HomeController : Controller
    {
        public IActionResult Index(Calculator calc)
        {
            switch (calc.operation)
            {
                case "+":
                    calc.Result = calc.Number1 + calc.Number2;
                    break;

                case "-":
                    calc.Result = calc.Number1 - calc.Number2;
                    break;

                case "*":
                    calc.Result = calc.Number1 * calc.Number2;
                    break;

                case "/":
                    if (calc.Number2 != 0)
                    {
                        calc.Result = calc.Number1 / calc.Number2;
                    }
                    break;
            }

            return View(calc);
        }
    }
}
```

Save the file using Ctrl + S.

5. Open the View Folder

In Solution Explorer, open Views → Home and double-click Index.cshtml. Delete the existing code and paste the following view code.

Index.cshtml — View Code

```
@model arithmeticmvc.Models.Calculator

<h2>Arithmetic operator</h2>

<form method="post">
    <label>Enter number 1</label>
```

```
    <input type="number" step="any" asp-for="Number1" />
    <br /><br />

    <label>Enter Number 2</label>
    <input type="number" step="any" asp-for="Number2" />
    <br /><br />

    <label>Select operation</label>
    <select asp-for="operation">
        <option value="+">Addition</option>
        <option value="-">Subtraction</option>
        <option value="*">Multiplication</option>
        <option value="/">Division</option>
    </select>

    <br /><br />
    <input type="submit" value="Calculate" />
</form>

@if (Model != null)
{
    <h3>Result = @Model.Result</h3>
}
```

6. Run the Application

1. Save all files.
2. Click the Run button or press F5.
3. Enter the first and second numbers.
4. Select Addition, Subtraction, Multiplication, or Division.
5. Click Calculate to display the result.

Expected Output

The page displays an arithmetic calculator with two number fields, an operation dropdown, and a Calculate button. After calculation, the result appears below the form.

Original Practical Screenshot

Practical 5A: Create an MVC Application to perform basic Arithmetic Operations.

Steps

1: Open visual studio 2022/2026

2: Create new project

- Click on create a new project
- Search for MVC
- Select project type as ASP.NET Core Web App(Model-View-controller)
- Then click next

2.2 Name the project arithmeticmvc->Click Next->Select Framework (latest available)

3. **Create Model File**

- In solution Explorer (If this window is not seen, go to view Menu and select it.)
- Right-click: Models →Add→Class
- Enter File name as Calculator.cs and click Add

```
namespace arithmeticmvc.Models{
public class Calculator{
public double Number1{ get;set;}
public double Number2{ get;set;}
public required string operation { get; set; }
public double Result { get; set; }
```

Press Ctrl+ S to save

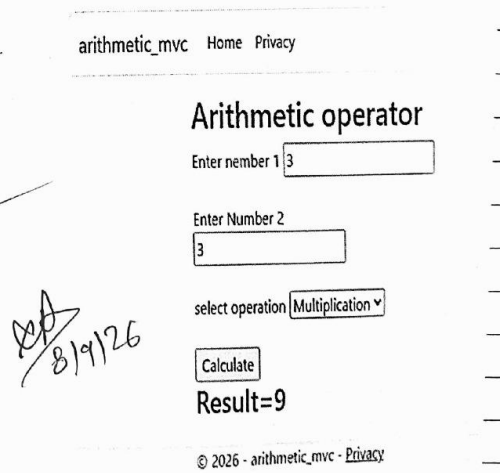
4. Open Controller

In solution explorer: open Controllers => Double click HomeController.cs

Modify Controller code:

Remove existing code inside HomeController and replace with:

```
using arithmeticmvc.Models;
using Microsoft.AspNetCore.Mvc;
using System.Diagnostics;
namespace arithmeticmvc.Controllers{
public class HomeController : Controller
{ public IActionResult Index(Calculator calc){
switch (calc.operation){
case "+":
calc.Result = calc.Number1 + calc.Number2;
break;
case "-":
calc.Result = calc.Number1 - calc.Number2;
break;
case "*":
calc.Result = calc.Number1 * calc.Number2;
break;
case "/":
if (calc.Number2 != 0)
{
calc.Result = calc.Number1 / calc.Number2;
}
break;} return View(calc);}}}
```



Save the file

5. Open views Folder

In solution explorer: Open: => Views -> Home

Double-click :=> Index.cshtml

11: Replace Existing View Code: Delete all existing code and paste:

Additional View Code (as shown on the second image)

The second image continues the Index.cshtml view code for Practical 5A. The code is included above in a clean, editable format. The original second page also begins Practical 5B, which has been excluded as requested.